

Student Worksheet

Can you train a tiny AI?

In this activity you will build, train, test, retrain, and analyze a tiny language model named Kairo.

Name: _____

Date: _____

Challenge 1 — Build it

My dataset

Dataset name or file:

What kind of text is in the dataset?

Prediction (before training)

What kind of text do you think the model will generate?

Which words or style do you expect to appear?

Challenge 2 — Train it

Loss notes

Starting train loss:

Ending train loss:

What happened to the loss value while training?

Reflection

Why do you think the loss changed?

Does lower loss mean the model understands like a person?

Challenge 3 — Talk to it

My prompt

Top predicted next token

Generated output

Reflection

Did the generated text make sense? Why or why not?

Which parts sounded fluent but might not be true?

Challenge 4 — Retrain it

Output Before Retrain

Prompt used:

Output recorded:

Example prompt:

Captain Rowan looked at the stars

Example observation before retrain:

The output sounds like a space story.

Output After Retrain

Prompt used (same prompt recommended):

Output recorded:

Example observation after pirate retrain:

The output includes pirate-style words such as "matey", "sail", or "tide".

What Changed and Why?

Compare the two outputs. Note style, vocabulary, repetition, and topic shifts.

Evidence	Before	After
Vocabulary		
Tone		
Repetition		
Topic		

Why do you think retraining changed the output?

Reflection

How did this change your understanding of how AI models learn?

Extension exercises

- Try two different datasets with the same prompt and compare outputs.
- Run one additional retrain round and record whether style shifts become stronger.
- Change max_new_tokens, temperature, or top_k and describe the effect.
- Write one prediction about what might happen with a much larger dataset.

Notes:

Challenge 5 — Inspect attention

Token with strongest attention

Prompt:

Token inspected:

Strongest attended token:

Reflection

Does attention mean understanding? Explain your answer.

What can attention help you inspect?

Challenge 6 — Save experiment

Experiment folder name

What would you need to record so someone else could repeat your experiment?

Challenge 7 — Understand it

What surprised you most?

What did you learn about AI?

One sentence I can now explain:

Language models learn...

Before/after retrain prompt ideas

Captain Rowan:
The station broke
Rain taps softly
The robot opened the door
Raise the black sail

New sample datasets

Use one starter file from data/samples/:

- space_adventure.txt
- pirate_dialogue.txt
- sci_fi_micro_story.txt
- short_poems.txt
- nature_notes.txt
- robot_helper.txt

Final reflection

- What did Kairo help you understand about language models?
- What changed when the training data changed?
- Why does this show that AI output depends on data?